FEATURE IMPORTANCE REPORT

Customer Churn Prediction for ABC Communications Ltd

Machine Learning Internship Programme
AnalystLab Africa

Week 3 Assignment

Prepared by: Christian Agyapong

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1. Introduction

Feature importance analysis was performed to determine which customer characteristics contributed most strongly to the predictions made by the machine learning model.

Understanding feature importance is important because a machine learning model should not only produce predictions but should also provide useful insights that can support business decisions.

For this project, **XGBoost** was selected for feature importance analysis because it achieved the strongest overall performance among the four evaluated models.

The XGBoost model achieved:

- **Accuracy:** 80.27%
- **Precision:** 66.11%
- **Recall:** 52.67%
- **F1-score:** 58.63%
- **ROC-AUC:** 84.46%

The feature importance analysis therefore focuses on identifying the variables that contributed most to the model's churn predictions.

2. Objective of Feature Importance Analysis

The objectives of this analysis were to:

1. Identify the most influential predictors of customer churn.
2. Rank the processed features according to their importance.
3. Understand the characteristics associated with churn prediction.
4. Translate technical model results into business insights.
5. Identify areas where ABC Communications Ltd. could focus on customer-retention efforts.

3. Evaluation Metrics

The feature importance values were obtained from the trained **XGBoost classification model**.

The processed dataset contained **50 features** after:

- Removing customerID
- Converting TotalCharges
- Handling missing values
- Engineering new variables
- Encoding categorical variables
- Scaling numerical variables

The categorical variables were transformed into encoded features such as:

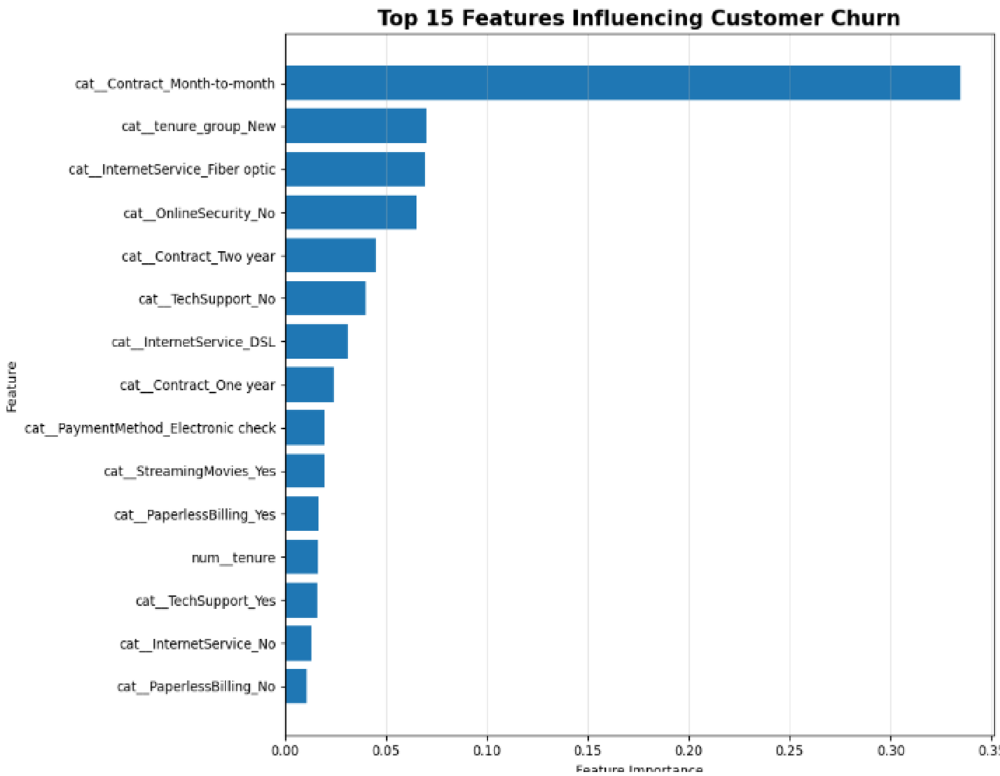
- Contract_Month-to-month
- Contract_One year
- Contract_Two year
- InternetService_Fiber optic
- OnlineSecurity_No
- PaymentMethod_Electronic check

The numerical variables included features such as:

- tenure
- MonthlyCharges
- TotalCharges
- SeniorCitizen
- total_services

6. Top Feature Importance Results

The XGBoost model identified the following features as the most important predictors:



6. Most Important Predictor

Contract – Month-to-month

with an importance value of approximately **0.3347**.

This feature was substantially more influential than the other predictors.

This indicates that the customer's contract arrangement played a major role in the XGBoost model's ability to distinguish between churn and non-churn customers.

Business interpretation

Customers with month-to-month contracts have greater flexibility to leave the company compared with customers who have longer-term contractual commitments.

ABC Communications Ltd. should therefore pay particular attention to customers who:

- Are on month-to-month contracts
- Have short tenure
- Have relatively high charges
- Do not use support-related services

These customers could be prioritized for proactive retention campaigns.

Important: Feature importance indicates how much a feature contributes to model predictions; it does **not** prove that the feature directly causes churn.

7. Tenure Group – New

The second most important feature was:

Tenure Group – New

with an importance of approximately **0.0700**.

This suggests that customers in the early stages of their relationship with the company are particularly important to the churn prediction model.

Business interpretation

New customers may not yet have developed strong loyalty to the company.

ABC Communications could introduce an enhanced onboarding strategy that includes:

- Welcome communication
- Early satisfaction surveys
- Technical support
- Personalized service recommendations
- New-customer incentives

8. Fiber Optic Internet Service

The third most important feature was:

Internet Service – Fiber optic

with an importance of approximately **0.0695**.

This indicates that internet service type contributes significantly to churn prediction.

Business interpretation

The company should investigate whether fiber-optic customers experience specific issues involving:

- Pricing
- Installation
- Service reliability
- Network performance
- Technical support
- Customer expectations

The model does not establish that fiber optic service causes churn. Instead, it identifies the feature as useful for distinguishing customer churn behavior.

9. Online Security

The feature:

Online Security – No

had an importance of approximately **0.0652**.

This suggests that online security subscription status contains useful information for predicting churn.

Business interpretation

Customers who do not subscribe to online security services may represent a different customer segment with different service needs or engagement patterns.

The company could investigate whether bundling security services with internet packages could improve customer engagement and retention.

However, any such strategy should be validated using controlled business experiments rather than assuming that adding the service will automatically reduce churn.

10. Contract Duration

Contract-related variables were among the most important features in the model.

The importance values were:

Contract Feature	Importance
Month-to-month	**0.334742**
Two year	0.045047
One year	0.024062

The combined importance of these contract-related features demonstrates that **contract structure is a major component of the model's churn predictions**.

Business implication

The company could investigate strategies such as:

- Loyalty programmes
- Long-term contract incentives
- Contract renewal campaigns
- Personalized offers
- Discounts for customers willing to commit for longer periods

These strategies should be designed carefully to avoid negatively affecting customer satisfaction.

11. Technical Support

The confusion matrices reveal an important difference between the models.

Two technical-support features appeared among the important predictors:

Feature	Importance
Tech Support – No	0.039910
Tech Support – Yes	0.015728

The relatively high importance of technical support status suggests that customer support engagement may be relevant to churn prediction.

Business interpretation

ABC Communications should examine whether customers experiencing unresolved technical problems are more likely to leave.

Potential improvements include:

- Faster issue resolution
- Proactive technical assistance
- Customer support follow-ups
- Better escalation procedures
- Monitoring repeated support requests

12. Payment Method

The feature:

Payment Method – Electronic check

had an importance value of approximately **0.0195**.

Payment method may capture differences in customer behavior, payment preferences, or customer segments.

The company could investigate whether customers using particular payment methods have different retention patterns.

13. Streaming and Billing Features

Other features appearing in the top 20 included:

- Streaming Movies – Yes
- Paperless Billing – Yes
- Paperless Billing – No
- Multiple Lines – Yes
- Multiple Lines – No

These features had smaller individual values compared with contract type and tenure-related variables.

Nevertheless, they contribute to the overall predictive capability of the model.

14. Numerical Feature Importance

Several numerical variables also contributed to the model.

Numerical Feature	Importance
Tenure	0.016397
Monthly Charges	0.009698
Total Charges	0.009391

Tenure

Tenure was more influential than both MonthlyCharges and TotalCharges.

This supports the importance of customer relationship duration in churn prediction.

Monthly Charges

Monthly charges contributed approximately **0.0097**.

This suggests that the customer's recurring financial commitment contains some predictive information.

Total Charges

Total charges contributed approximately **0.0094**.

Although less important than tenure, this feature provides information about the customer's cumulative financial relationship with the company.

15. Business Insights from Feature Importance

The feature-important analysis suggests several important business patterns.

Insight 1: Contract structure matters

Contract type is the strongest group of predictors, particularly month-to-month contracts.

Insight 2: Early customer relationships require attention

New customers appear more vulnerable or distinguishable in terms of churn risk.

Insight 3: Service type matters

Internet service type, particularly fiber optic service, contributes significantly to predictions.

Insight 4: Customer support may influence retention

Technical support and online security features are among the important predictors.

Insight 5: Customer segmentation can improve retention

The company can use these characteristics to create targeted retention strategies rather than treating every customer identically.

16. Feature Importance and Business Action

Important Feature	Business Interpretation	Possible Action
Month-to-month contract	Higher flexibility	Targeted retention offers
New tenure group	Early customer relationship	Improved onboarding
Fiber optic	Important service segment	Investigate service experience
Online Security – No	Different service profile	Consider relevant service bundles
Tech Support – No	Potential support gap	Proactive technical support
Two-year contract	Longer commitment	Study characteristics of loyal customers
DSL	Important customer segment	Investigate service satisfaction
Electronic check	Distinct payment segment	Analyze payment/customer behavior
Streaming Movies	Customer service profile	Segment customer offers
Monthly Charges	Financial commitment	Evaluate pricing sensitivity

17. Limitations of Feature Importance

Feature importance should be interpreted carefully.

16.1 Importance is not causation

A high feature importance does not mean that the feature directly causes customer churn.

For example, the model identifies month-to-month contracts as highly important, but this does not prove that changing a customer's contract automatically prevents churn.

16.2 Correlated features

Some variables may contain overlapping information.

For example:

- Tenure
- TotalCharges
- Contract type

may be related to the customer's overall relationship with the company.

16.3 Model-specific interpretation

The feature importance values presented in this report come from XGBoost and therefore describe how that particular model uses the available features.

18. Recommendations

Based on the feature importance analysis, ABC Communications Ltd. should consider:

1. Create a high-risk customer segment

Prioritize customers with combinations of characteristics associated with high predicted churn risk.

2. Focus on month-to-month customers

Develop targeted retention strategies for customers without long-term contracts.

3. Strengthen new-customer onboarding

Provide additional support during the early stages of the customer relationship.

4. Investigate fiber-optic customer experience

Analyze complaints, service interruptions, pricing, and support satisfaction among fiber-optic customers.

5. Improve technical support

Identify customers experiencing repeated technical problems and proactively intervene.

6. Use explainable AI techniques

Future versions should use tools such as **SHAP** to provide more detailed explanations of individual customer predictions.

Conclusion

The feature importance analysis provided valuable insight into the factors used by the XGBoost model to predict customer churn.

The strongest predictor was **Contract – Month-to-month**, with an importance of approximately **0.3347**, followed by **Tenure Group – New**, **Fiber Optic Internet Service**, and **Online Security – No**.

The results indicate that customer contract structure, tenure, service type, and support-related characteristics are particularly useful for predicting churn.

These findings can help ABC Communications Ltd. move beyond simply predicting churn and begin developing targeted customer-retention strategies.

However, feature importance should be interpreted as **predictive evidence rather than causal evidence**. Additional statistical analysis, explainability techniques, and controlled business experiments should be used before implementing major customer-retention decisions.